

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent Application Publication	20250282533
Kind Code	A1
Publication Date	September 11, 2025
Inventor(s)	Gary; Lamaun T.

SMART MOVING CONTAINER

Abstract

A smart moving container, system, and method for tracking the contents within the smart moving container and the location thereof also collectively monitors and/or mitigates factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management. The smart moving container has location tracking devices, temperature sensors, weight sensors, impact sensors, humidity sensors, and inventory identification codes for real-time tracking and monitoring. The smart moving container has specialized padding for content protection.

Inventors:	Gary; Lamaun T. (Schenectady, NY)
Applicant:	Gary; Lamaun T. (Schenectady, NY)
Family ID:	96948558
Appl. No.:	19/186084
Filed:	April 22, 2025

Related U.S. Application Data

us-provisional-application US 63562969 20240308

Publication Classification

Int. Cl.: B65D51/24 (20060101); B65D43/16 (20060101)

U.S. Cl.:

CPC B65D51/248 (20130101); B65D43/16 (20130101); B65D2203/10 (20130101)

Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION [0001] This application claims the benefit of priority of U.S. provisional application No. 63/562,969, filed Mar. 8, 2024, the contents of which are herein incorporated by reference.

BACKGROUND OF THE INVENTION

[0002] The present invention relates to moving containers and, more particularly, to a smart moving container for tracking the contents therein and the location thereof.

[0003] A variety of different types of containers are available for moving items and their general storage. These containers may be intended for general, residential and/or commercial use during the transportation of numerous types of items (e.g., contents of a home, commercial commerce, etc.) using a variety of transportation means (e.g., trucks, airplanes, cars, trains, etc.). Rigid containers are typically preferable as they are durable, can maintain their shape, and thereby protect the stored items from the elements and damage. Another advantage is that the rigid containers are usually easily washable and, therefore, reusable. In addition to protecting the contents of such containers it is also critical to track their whereabouts during transport.

[0004] Conventional package tracking is well-known in the art, being used by postal services, couriers, and other shipping and delivery services to determine the location of a package. A package, letter, or box is assigned a tracking number or code when it is first given by a sender to the delivery service. Typically, a barcode is affixed to the package and may be scanned with an optical reader. Alternatively, other technology such as magnetic strips, Radio-frequency identification (RFID) or Near-field communication (NFC) tags may be used. At key points in the package's journey to its destination, the package is scanned, with the location and progress of the package being made available to anyone with the barcode. For example, a standard tracking system in the shipping industry has been the utilization of barcode scanning technology to identify and process packages as they are transported around the world.

[0005] Of course, location and damage prevention are only a few of the concerns when transporting goods. Other factors that impact the transportation of goods may be related to weight, temperature, humidity, impact detection, and inventory management. None of these factors are a mystery to transportation experts and each may be mitigated by well-known devices and/or techniques, however, the transportation industry appears to be lacking a single moving container that will monitor and/or mitigate all of these factors in an integrated single unit and design.

[0006] As can be seen, there is a need for an improved moving container that facilitates tracking the contents therein and the location thereof but also monitors and/or mitigates factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management.

SUMMARY OF THE INVENTION

[0007] The present invention is directed to a smart moving container that is an integrated single unit design that facilitates tracking the contents therein and the location thereof and also monitors and/or mitigates factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management.

[0008] In one aspect of the present invention, a smart moving container is provided for tracking the contents therein and the location thereof that also monitors and/or mitigates factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management. The smart moving container comprises a container body comprising a plurality of sidewalls forming an interior and exterior wall surface and a body cavity between a base of the container body and the plurality of sidewalls; a container lid comprising one or more portions; a connection system for connecting the container lid to the container body; one or

more location tracking devices capable of tracking a real-time location of the smart moving container; one or more temperature sensors capable of monitoring and transmitting a real-time temperature of contents within the body cavity of the container body; one or more weight sensors capable of monitoring and transmitting a real-time weight of the smart moving container; one or more impact sensors capable of monitoring for and transmitting an alert when a real-time impact or shock is applied to the smart moving container; one or more humidity sensors capable of monitoring and transmitting a real-time humidity exposure level for the smart moving container; one or more inventory identification codes; and specialized padding capable of protecting the contents within the body cavity of the container body.

[0009] In another aspect of the present invention, a method is provided for using a smart moving container for tracking the contents therein and the location thereof as well as monitors and/or mitigates factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management. The method comprising (i) attaching one or more location tracking devices capable of tracking a real-time location of the smart moving container; (ii) attaching one or more temperature sensors capable of monitoring and transmitting a real-time temperature of contents within the smart moving container; (iii) attaching one or more weight sensors capable of monitoring and transmitting a real-time weight of the smart moving container; (iv) attaching one or more impact sensors capable of monitoring for and transmitting an alert when a real-time impact or shock is applied to the smart moving container; (v) attaching one or more humidity sensors capable of monitoring and transmitting a real-time humidity exposure level for the smart moving container; (vi) applying one or more inventory identification codes to the smart moving container; (vii) applying a specialized padding to the moving container; (viii) assigning a unique inventory tracking identification code to the smart moving container; (viii) monitoring a location, temperature, weight, impact, humidity and inventory status of the smart moving container using the one or more location tracking devices, the one or more temperature sensors, the one or more humidity sensor, and the inventory tracking identification code; and (ix) transmitting, the location, temperature, weight, impact, humidity and inventory status of the smart moving container to at least one user device.

[0010] In a third aspect, a user device is provided and configured for interfacing with the smart moving container for tracking the contents therein and the location thereof as well as monitoring and/or mitigating factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management.

[0011] In a fourth aspect, a smart moving container application (alternatively referred to herein as an “app”) may be executed for tracking the contents therein and the location thereof as well as monitoring and/or mitigating factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management.

[0012] In another aspect, the container body and the container lid are formed from a high-density (HD) polyethylene material. In some embodiments, various components of the smart moving container may be manufactured of polypropylene (PP), such as a sensor casing.

[0013] In another aspect, the container lid is connectable to the container body through the connection system to form a fixed connection between the container lid and the container body.

[0014] In another aspect, the fixed connection between the container lid and the container body is airtight and liquid tight.

[0015] In another aspect, the container lid comprises a first hinged portion and a second hinged portion.

[0016] In another aspect, the connection system comprises one or more hinges for connecting the one or more portions of the container lid to the container body.

[0017] In another aspect, the connection system for connecting the container lid to the container body comprises a press-fit design.

[0018] In another aspect, the container body comprises one or more handles for carrying the smart

moving container.

[0019] In another aspect, the one or more tracking devices include at least one of a radio-frequency identification (RFID) tag or a barcode.

[0020] In another aspect, the one or more location tracking devices, the one or more temperature sensors, the one or more weight sensors, the one or more impact sensors, and the one or more humidity sensors are located within the body cavity.

[0021] In another aspect, at least one of the one or more location tracking devices, the one or more temperature sensors, the one or more weight sensors, the one or more impact sensors, and the one or more humidity sensors are located within the body cavity is mounted on a particular one sidewall of the plurality of sidewalls. In some embodiments, the one or more weight sensors may be located at the bottom of the container.

[0022] In another aspect, the container lid and the container body each contain a sealing member providing an airtight and/or liquid-tight seal therebetween when the container lid and the container body are fully pressed together.

[0023] In another aspect, a first portion of the container lid is hingedly connected to a particular one side wall of the plurality of sidewalls, and a second portion of the container lid is hingedly connected to a particular another one sidewall of the plurality of sidewalls.

[0024] In another aspect, the container lid allows for the smart moving container to be stacked.

[0025] In another aspect, the smart moving container comprises a first set of LED lights configured in a star pattern on the interior surface of a first one sidewall of the plurality of sidewalls; and a second set of LED lights configured in a star pattern on the interior surface of a second one sidewall of the plurality of sidewalls.

[0026] The methods and systems described herein can be implemented by data processing systems, such as one or more smartphones, tablet computers, desktop computers, laptop computers, smart watches, wearable, audio accessories, on-board computer, and other user devices and consumer electronic devices. The methods and systems described herein can also be implemented by one or more data processing systems which execute executable computer program instructions, stored in one or more non-transitory machine-readable media that cause the one or more data processing systems to perform the one or more methods described herein when the program instructions are executed. Thus, the embodiments described herein can include methods, data processing systems, and non-transitory machine-readable media.

[0027] The above summary does not include an exhaustive list of all embodiments in this disclosure. All systems and methods can be practiced from all suitable combinations of the various aspects and embodiments summarized above, and also those disclosed in the detailed description below.

[0028] These and other features, aspects and advantages of the present invention will become better understood with reference to the following drawings, description, and claims.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

[0029] The preferred embodiments of the invention will hereinafter be described in conjunction with the appended drawings provided to illustrate and not to limit the invention, where like designations denote like elements, and in which:

[0030] FIG. 1 presents a top view of a smart moving container in accordance with an embodiment of the present invention, shown in an open position;

[0031] FIG. 2 presents a front perspective view thereof, shown in a partially closed position;

[0032] FIG. 3 presents a side perspective view thereof, shown in a fully closed position;

[0033] FIG. 4 presents a perspective view thereof, shown in a partially open position exposing the

contents therein;

[0034] FIG. 5 presents a diagram of a cloud network services architecture in accordance with an embodiment of the present invention, including the smart moving container of FIG. 1;

[0035] FIG. 6 presents an illustrative data processing system in accordance with an embodiment of the present invention, the system being configured for use with the smart moving container of FIG. 1;

[0036] FIG. 7 presents an illustrative architecture for a smart moving container app in accordance with an embodiment of the present invention;

[0037] FIG. 8 presents a flowchart illustrative of operations for user onboarding and subscription in accordance with an embodiment of the present invention;

[0038] FIG. 9 is a schematic view of an illustrative user profile for use with the smart moving container of FIG. 1; and

[0039] FIG. 10 presents a flowchart of a method of tracking the contents of and the location of a smart moving container as well as monitoring and/or mitigating factors that impact the transportation of goods in accordance with an embodiment of the present invention.

[0040] Like reference numerals refer to like parts throughout the several views of the drawings.

DETAILED DESCRIPTION OF THE INVENTION

[0041] The following detailed description is of the best currently contemplated modes of carrying out exemplary embodiments of the invention. The description is not to be taken in a limiting sense but is made merely for the purpose of illustrating the general principles of the invention, since the scope of the invention is best defined by the appended claims.

[0042] As used herein, the word “exemplary” or “illustrative” means “serving as an example, instance, or illustration.” Any implementation described herein as “exemplary” or “illustrative” is not necessarily to be construed as preferred or advantageous over other implementations. All of the implementations described below are exemplary implementations provided to enable persons skilled in the art to make or use the embodiments of the disclosure and are not intended to limit the scope of the disclosure, which is defined by the claims. Furthermore, there is no intention to be bound by any expressed or implied theory presented in the preceding technical field, background, brief summary or the following detailed description. It is also to be understood that the specific devices and processes illustrated in the attached drawings, and described in the following specification, are simply exemplary embodiments of the inventive concepts defined in the appended claims. Hence, specific dimensions and other physical characteristics relating to the embodiments disclosed herein are not to be considered as limiting, unless the claims expressly state otherwise.

[0043] The term “smart” moving container as used herein refers to a moving container that incorporates electronic sensors, networked communication capabilities, and automated monitoring functionalities.

[0044] Broadly, one embodiment of the present invention is a smart moving container, system, and method for tracking the contents therein and the location thereof and also collectively monitors and/or mitigates factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management. That is, in accordance with the principles of the disclosed embodiments, a smart moving container that is a single fully integrated container and design is configured with one or more location tracking devices capable of tracking a real-time location of the smart moving container, one or more temperature sensors capable of monitoring and transmitting a real-time temperature of contents within the body cavity of the container body, one or more weight sensors capable of monitoring and transmitting a real-time weight of the smart moving container, one or more impact sensors capable of monitoring for and transmitting an alert when a real-time impact or shock is applied to the smart moving container, one or more humidity sensors capable of monitoring and transmitting a real-time humidity exposure level for the smart moving container, one or more non-tamperable inventory identification

codes, and specialized padding capable of protecting the contents within the body cavity of the container body. As such, the disclosed embodiments address problems with moving containers related to actively and effectively monitoring and/or mitigating factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management. The tracking and monitoring processes are further advantaged using a smart moving container app executing on a data processing system that is in communication with the smart moving container before, during and/or after transit. As result, the smart moving container, system and method of the disclosed embodiments provide an advantageous improvement of practical applications that include transportation, commerce shipping and moving techniques, and moving container tracking and monitoring.

[0045] Alert algorithms may be used to ensure the timely notification of potential issues during the transportation of goods. These algorithms can include threshold-based alerts, which trigger notifications when sensor readings, such as temperature or humidity, exceed predefined limits. Specific events or combinations of events may trigger alerts in some cases. Additionally, duration-based alerts may be activated if a condition persists beyond an acceptable timeframe, while rate-of-change alerts respond to rapid fluctuations in sensor data. Geofencing alerts may activate when the container enters or exits specified geographic zones. Combined logic alerts may be utilized, e.g., using Boolean operations to evaluate multiple conditions simultaneously, and state-based alerts, which respond to changes in the device's operational state. For advanced monitoring, anomaly detection algorithms may leverage machine learning models trained on historical data to identify deviations from expected patterns. Alerts can also be triggered based on time-series analysis, indicating sudden changes in trends or when data points deviate from expected patterns.

[0046] In some embodiments, the container may be fitted with scale flex flooring positioned above a weight sensor(s), enabling the weight sensor(s) to monitor the overall weight of the container contents.

[0047] In some embodiments, the weight sensor(s) may be housed within a structural case resting in the body cavity.

[0048] In some embodiments, the temperature and humidity sensor(s) may be provided as a module and may be housed within a sensor casing.

[0049] In some embodiments, the container may have a printed circuit board assembly (PCBA) main controller operative to integrate signals from various sensors and to wirelessly transmit the integrated signals to a processor. The PCBA may be housed within an enclosure inside the container, using vibration-dampening standoffs or grommets. The PCBA may be configured to ensure signal integrity, minimize noise, and efficiently manage power distribution. In some cases, tamper detection sensors may be used to determine if the electronics enclosure has been opened.

[0050] In embodiments, the smart moving container may be powered by a rechargeable Lithium-ion (Li-ion) or Lithium-polymer (LiPo) battery pack. These power sources offer good energy density and a reasonable lifespan, e.g., about 2-6 months or more. The battery pack may be recharged, for example, via a USB-C interface integrated into an electronics module housing, accessible from the outside or from the inside of the container. Alternatively, the battery pack may be wirelessly charged. Docking stations may be provided in some cases for bulk charging of multiple smart moving containers in a warehouse environment. The container may have a power saving mode with lower power consumption to save battery life.

[0051] The container body and lid may be manufactured using high-density polyethylene (HDPE) or polypropylene through injection molding to provide a robust and lightweight structure. A molded cavity or a dedicated double-wall space may be incorporated to securely house the electronics module in a rugged, sealed enclosure made from durable materials, such as ABS, polycarbonate, or metal, with gaskets or seals to achieve a high Ingress Protection (IP) rating, such as IP 65 or IP 67, to prevent dust and water ingress, ensuring protection against environmental factors. The electronics module may contain the printed circuit board assembly (PCBA), battery,

and antennas.

[0052] The sensors, including temperature, humidity, weight, and impact sensors, may be connected via internal wiring to the central processing unit, which processes and transmits data to external devices. This wiring is generally insulated and secured to prevent interference with the container's protective features. Temperature and humidity sensors may be placed within the lid, protected by a sealing mechanism that allows these sensors to function without exposure to external elements. The weight sensor may be installed on a sturdy platform at the bottom of the container, ensuring even weight distribution for precise measurements.

[0053] The smart moving container system may employ several security protocols to ensure the integrity and confidentiality of data collected and transmitted during the transportation process. Encryption protocols such as Transport Layer Security (TLS) or Secure Sockets Layer (SSL) are utilized to secure data transmission between the container and external systems, preventing unauthorized interception. Strong authentication mechanisms, including multi-factor authentication (MFA) and digital certificates, verify the identity of users and devices accessing the system, ensuring that only authorized personnel can interact with sensitive information. Access control measures further restrict data access based on user roles and permissions, safeguarding sensitive data from unauthorized access. Data integrity checks using hashing algorithms are implemented to verify that data has not been altered or tampered with during transmission. Additionally, secure Application Programming Interfaces (APIs) are employed for interactions with other applications or services, incorporating authentication and encryption to protect data exchanges. Regular security audits and vulnerability assessments are conducted to identify and mitigate potential security risks, ensuring the system remains robust against emerging threats. Secure Boot ensures the device only boots with authenticated firmware, kept up to date with secure, cryptographically signed, Over-The-Air (OTA) updates.

[0054] The smart moving container may communicate with user devices and existing inventory management systems through a combination of wireless technologies and protocols designed for efficient data transmission and integration. The container may be equipped with communication modules that utilize Wi-Fi, Bluetooth, or cellular networks to establish a connection with user devices such as smartphones, tablets, or computers. This connectivity allows for real-time updates and notifications regarding the container's status, location, and environmental conditions. Additionally, the container may employ Internet of Things (IoT) protocols like MQTT (Message Queuing Telemetry Transport) or CoAP (Constrained Application Protocol) to facilitate low-bandwidth, high-latency communication of data such as sensor readings, GPS coordinates, battery status, and alerts. The container may use Application Programming Interfaces (APIs) to send and receive data, such as inventory levels and tracking information, ensuring synchronization with existing inventory management systems. This integration may be enhanced by cloud connectivity, allowing for centralized monitoring and management.

[0055] Referring to FIGS. 1-10, FIGS. 1-4 depict various views of illustrative smart moving container **100**. As shown, the smart moving container comprises a container body **180** comprising a plurality of sidewalls **120** forming an interior wall surface and an exterior wall surface and a body cavity **130** between a base **110** of the container body **180** and the plurality of sidewalls **120**. The container body **180** is illustrated as being generally rectangular shaped with four sidewalls, however, the container body **180** may have other shapes and/or a different number of sidewalls. FIG. 1 shows the smart moving container **100** in a fully open position thereby exposing the contents and other components configured therein. The smart moving container **100** may be designed and manufactured in a wide variety of sizes to accommodate various shipping and transportation needs. The smart moving container **100** further comprises a container lid **105** comprising one or more portions. That is, the container lid **105** may have multiple portions as shown in FIG. 1, for example, or may be constructed as a single portion or body and may be configured such that multiple smart moving containers **100** may be stacked one on the other. The

container lid **105** is physically connected with the container body **180** through a connection system **115** (e.g., one or more hinges or any other type of suitable connection component). For example, one or more hinges can be formed such that the hinge is permanently and/or securely connected to the container lid **105** and the container body **180** and would be required to be cut or otherwise damaged to separate the container lid **105** from the container body **180**. Thus, a first portion of the container lid **105** may be hingedly connected to a particular one sidewall **120** and a second portion of the container lid **105** may be hingedly connected to a particular another one sidewall **120** of the plurality of sidewalls. As such, the container lid **105** is connectable to the container body **180** through the connection system **115** to form a fixed connection between the container lid **105** and the container body **180** such that the fixed connection therebetween is airtight and liquid-tight when in a closed position, thereby protecting contents **135** inside the smart moving container **100**. In an embodiment, the connection system **115** may have a press-fit design between the container lid **105** and the container body **180** that is formed using one or more channels or grooves on the bottom surface of the container lid **105** and the top surface of the container body **180**. Alternatively, the container lid **105** and the container body **180** may each contain a sealing member providing an airtight and/or liquid-tight seal therebetween when the container lid **105** and the container body **180** are fully pressed together. In an embodiment, the container body **180** and/or the container lid **105** are manufactured from a high-density (HD) polyethylene material given this material's generally favorable shock absorbing qualities. The container lid **105** may also provide for a fully closed and locked position when engaged with the container body **180**. The container lid **105** and/or the container body **180** may optionally contain one or more support ribs on their surfaces in order to promote the strength and stability of the smart moving container **100**. Further, the container body **180** may include one or more handles **125** for carrying or otherwise transporting the smart moving container **100**. In another aspect, the smart moving container comprises a first set of light-emitting diode (LED) lights **182** configured in a star pattern on the interior surface of a first one sidewall of the plurality of sidewalls; and a second set of LED lights **184** configured in a star pattern on the interior surface of a second one sidewall of the plurality of sidewalls. In this way, the resident light sources may be used to illuminate the body cavity **130**, for example, thereby providing the user with improved visuals in low light settings such as residential homes, storage facilities, offices, warehouses, trucks, vans, and the like. The star formation will facilitate delivering the light source throughout the body cavity **130** area (e.g., all the corners and crevices) when the user is browsing, packing and/or unpacking the contents **135**. Of course, the star formation is one of many different lighting configurations that may be utilized for illuminating the body cavity **130** and the contents therein.

[0056] Importantly, the smart moving container **100** is configured with a number of sensors and other components that enable the real-time monitoring and/or mitigation of factors that impact the transportation of goods specific to damage control, weight, temperature, humidity, impact detection, and inventory management. In particular, one or more location tracking devices **140** capable of tracking a real-time location of the smart moving container **100** are affixed to a particular portion or area of the smart moving container **100** (e.g., the underside of the lids **105**). For example, the one or more location tracking devices **140** may be a radio-frequency identification (RFID) tag or other similar wireless tracking device (e.g., an Apple AirTag®, which uses Bluetooth® short-range wireless technology and Ultra-Wideband (UWB) technology to track and locate items). One or more temperature sensors **145** are provided (e.g., mounted on an inside surface of one of the sidewalls **165** of the container body **180**) for monitoring and transmitting a real-time temperature of contents **135** within the body cavity **130** of the container body **180**. Also, one or more weight sensors **155** may be provided for monitoring and transmitting a real-time weight of the smart moving container **100**. This will trigger an alert, for example, if there is a substantial change in the reported real-time weight thereby indicating some change to either the contents **135** (e.g., theft of the contents **135**) or the smart moving container **100** itself (e.g., damage

and material loss). These weights measurements may also be used for calculating overall transportation costs associated with shipping one or more of the smart moving containers **100**. One or more impact sensors **160** may be provided for monitoring and transmitting an alert when a real-time impact or shock is applied to the smart moving container **100** during transit or otherwise. Further, one or more humidity sensors **170** may be provided for monitoring and transmitting a real-time humidity exposure level for the smart moving container **100**. In this way, an assessment may be made with respect to any damage of the contents **135** that may be associated with extreme or harmful humidity levels. One or more inventory identification codes **175** (e.g., a barcode) may be provided for inventorying the contents **135** of each smart moving container **100** before, during, and/or after shipment, and specialized padding **150** may be provided for protecting the contents **135** within the body cavity **130** of the container body **180** before, during and/or after shipment. Each of the aforementioned sensors may be configured for transmitting their real-time data collection to one or more data processing units **300**, for example. Alternatively, the smart moving container **100** may be configured with a control unit that comprises, among other components, a communications subsystem that is communicatively coupled to such sensors and the communication subsystem transmits the real-time data collected.

[0057] Turning our attention to FIG. 5, a diagram of a cloud network services architecture **200** is shown for employing the smart moving container **100** for tracking the contents therein and the location thereof as well as for monitoring and/or mitigating factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management in accordance with an embodiment. The cloud network services architecture **200** may include network **205** connected with several data processing systems. For example, the user device/data processing system **300** (see also, FIG. 6) and server **260** through which a user may employ the smart moving container **100** for tracking the contents therein and the location thereof as well as for monitoring and/or mitigates factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management. For example, graphical user interface (GUI) **245** may display various real-time data including but not limited to an impact event **215**, a current weight **220**, a current humidity level **225**, and a current temperature **230**. The network **205** may include at least server(s), access point(s) and database(s). Cloud, cloud service, cloud server and cloud database are broad terms and are to be given their ordinary and customary meaning to one of ordinary skill in the art and include, without limitation, any database, data repository or storage media which store content typically associated with and managed by a user. A cloud service may include one or more cloud servers and cloud databases that provide for the remote storage of content as hosted by a third-party service provider or operator. A cloud server may include a Hypertext Transfer Protocol (HTTP)/Hypertext Transfer Protocol Secure (HTTPS) server sending and receiving messages in order to provide web-browsing interfaces to client web browsers as well as web services to send data to integrate with other interfaces (e.g., as executed on the data processing system **300**). The cloud server may be implemented in one or more servers and may send and receive content in various forms and formats, user supplied and/or created information/content and profile/configuration data that may be transferred from or stored in a cloud database. A cloud database may include one or more physical servers, databases or storage devices as dictated by the cloud service's storage requirements. The cloud database may further include one or more well-known databases (e.g., a Structured Query Language [SQL] database) or a fixed content storage system to store content, user profile information, configuration information, administration information and any other information necessary to execute the cloud service. In various embodiments, one or more networks providing computing infrastructure on behalf of one or more users may be referred to as a cloud, and resources may include, without limitation, data center resources, applications (e.g., software-as-a-service or platform-as-a-service) and management tools. The term "cloud computing" as referred to herein implies the on-demand availability of computer system resources, especially data

storage and computing power, without direct active management by the user. The term is generally used to describe data centers available to many users over the Internet. Large clouds, predominant today, often have functions distributed over multiple locations from central servers. If the connection to the user is relatively close, it may be designated an edge server. Clouds may be limited to a single organization (e.g., enterprise clouds), be available to many organizations (e.g., public cloud), or a combination of both (e.g., hybrid cloud). Cloud computing relies on sharing resources to achieve coherence and economies of scale. Advocates of public and hybrid clouds note that cloud computing allows companies to avoid or minimize up-front IT infrastructure costs. Proponents also claim that cloud computing allows enterprises to get their applications up and running faster, with improved manageability and less maintenance, and that it enables IT teams to more rapidly adjust resources to meet fluctuating and unpredictable demand. In an embodiment, the data processing system **300** (e.g., a smartphone) and/or server **260** communicate over communications links **235** for accessing each other and for communicating with the smart moving container **100** before, during, and after shipment. Such operations may be facilitated and managed by a smart moving container app **400** (see also, FIG. 7) executing on the data processing system **300** and/or the server **260**.

[0058] Turning our attention to FIG. 6, an illustrative data processing system **300** is shown configured for use with the smart moving container **100** in accordance with an embodiment. As will be appreciated, a “data processing system” in the context herein may comprise a wide variety of devices such as any type of hardware device, mobile devices, smartphones, laptop computers, desktop computers, tablets, servers, and wearable devices, to name just a few, that execute applications (e.g., a mobile application) in accordance with the principles of the disclosed embodiments herein. As shown, the data processing system **300** comprises processor **302** for executing program code (e.g., the smart moving container app **400**) and communications interface **314** for managing communications to and from the data processing system **300**, memory **306**, data storage **310**, and/or read-only memory (ROM) **308** for storing program code and data, and power source **318** for powering the data processing system **300**. The memory **306** is coupled to the bus **304** for storing computer-readable instructions to be executed by the processor **302** (e.g., execution of the smart moving container app **400**). The processor **302** may include both general and special purpose microprocessors and may be the sole processor or one of multiple processors of the device. Further, the processor **302** may comprise one or more central processing units (CPUs) and may include, be supplemented by, or incorporated in, one or more application-specific integrated circuits (ASICs) and/or one or more field programmable gate arrays (FPGAs). The memory **306** may also be utilized for storing temporary variables or other intermediate information during the execution of the instructions by the processor **302**. The data storage device **310**, such as a magnetic, optical, or solid-state device may be coupled to the bus **304** for storing information and instructions for the processor **302** including, but not limited to, the smart moving container app **400**. Data storage device **310** and the memory **306** may each comprise a non-transitory computer readable storage medium and may each include high-speed random access memory, such as dynamic random access memory (DRAM), static random access memory (SRAM), double data rate synchronous dynamic random access memory (DDR RAM), or other random access solid state memory devices, and may include non-volatile memory, such as one or more magnetic disk storage devices such as internal hard disks and removable disks, magneto-optical disk storage devices, optical disk storage devices, flash memory devices, semiconductor memory devices, such as erasable programmable read-only memory (EPROM), electrically erasable programmable read-only memory (EEPROM), compact disc read-only memory (CD-ROM), digital versatile disc read-only memory (DVD-ROM) disks, or other non-volatile solid state storage devices.

[0059] Tag and sensor manager **326** is used to manage the various tags and sensors, as detailed above, of the smart moving container **100** in accordance with the disclosed embodiments. Location-based services manager **322** facilitates the delivery of location-based services (e.g.,

Global Position System (GPS) tracking) either independently or on the data processing system **300** using GPS transceiver **312** and antenna **330**. This allows the data processing system **300** to register the exact location of the smart moving container(s) **100** and/or the user of the data processing system **300**, for example, as the user roams from one location to another location such that the services offered via the smart moving container processing hereunder may be tailored to a current location and/or the needs of the user as they may change based on their current location which may influence their selection of the number of smart moving containers **100** needed or in providing certain specific transportation instructions, for example.

[0060] The input/output devices **316** may include peripherals, such as a biometric reader, an NFC device (e.g., NFC tag reader), camera, printer, scanner (e.g., a Quick Response code [QR] scanner), touchscreen display, etc. For example, the input/output devices **316** may include a display device such as a cathode ray tube (CRT), plasma monitor, liquid crystal display (LCD) monitor or organic light-emitting diode (OLED) monitor for displaying information to the user, a keyboard, and a pointing device such as a mouse or a trackball by which the user can provide input to the data processing system **300** or an associated display device **324**, for example, that may also be managed by graphical user interface generator **328**. The communications interface **314** is used to facilitate communications across various types of communications links within the cloud network services architecture **200**, for example. This may take the form, for example, of a wide area network connection that communicatively couples the data processing system **300** with the access points of the network **205** which may be a cellular communications service. Similarly, communications managed by the communications interface **314** may take the form, for example, of a local Wi-Fi network interface or Ethernet interface the communicatively couples the data processing system **300** with the well-known Internet, a local area network (LAN), and ultimately the data processing system **300**. In the instant embodiment, the smart moving container app **400** and/or the communications interface **314** may include a communications stack for facilitating communications over the respective communications link (e.g., the communications links **235**). The data processing system **300** may also communicate with other devices via a network (e.g., a wireless communications network) or communications protocol (e.g., Bluetooth®). For example, such communications interfaces may be a receiver, transceiver, or modem for exchanging wired or wireless communications in any number of well-known fashions. For example, the communications interface **314** may be an integrated services digital network (ISDN) card or modem/router used to facilitate data communications of various well-known types and formats. Further, illustratively, the communications interface **314** may be a LAN card used to provide data communication connectivity to a comparable LAN. As will be appreciated, the functionality of the communications interface **314** is to send and receive a variety of signals (e.g., electrical, optical, or other signals) that transmit data streams representing various data types.

[0061] Turning our attention to FIG. 7, an illustrative architecture for the operation of the smart moving container app **400** is presented in accordance with an embodiment. As will be appreciated, the architecture may be used, illustratively, in conjunction with the cloud network services architecture **200** and/or the data processing system **300** for launching and executing the smart moving container app **400** and its associated operations. As shown, the architecture for the operations of the smart moving container app **400** provides several interfaces and engines used to perform a variety of functions such as the collection, aggregation, manipulation, processing, analyzing, verification, authentication, and display of applicable real-time information and data that are useful to realize the delivery of the smart moving container operations of the disclosed embodiments. More particularly, data display interface module **418** and communications module **412** are used to facilitate the input/output and display of electronic data and other information to, illustratively, the users employing the data processing system **300** (e.g., a touch screen of the data processing system **300**) and executing the smart moving container app **400**. The data collection module **406** facilitates data gathering from the plurality of users and other third parties. The

location-based services module **420** provides for the delivery of location-based services in order for the geographic locations of the smart moving container(s) **100** and/or the users to be identified and displayed (e.g., GPS locations). The communications module **412** will also facilitate communications by and through the data processing system **300**, for example.

[0062] Execution engine **402** may be employed to deliver the smart moving container services herein through the execution of the smart moving container app **400**. In such delivery, the execution engine **402** will operate and execute, as further detailed herein below, with at least the following program modules: graphical user interface module **404**, data collection module **406**, user subscription module **408**, user profile module **410**, communications module **412**, smart moving container operations module **414**, RFID and tag and barcode administration and management module **416**, sensor administration and management module **422**, smart moving container inventory administration and management module **424**, third-party platform administration and management module **426**, and user device administration and management module **428**. The user subscription module **408** provides for the delivery and management of a subscription-based smart moving container services model whereby individual users may subscribe in order to access the smart moving container services through execution of the smart moving container app **400**. In an embodiment, the user may initially subscribe, as facilitated by the user subscription module **408** for a defined fee in a tier subscription system such that users who pay more for their access subscription are provided priority and other advantages over other users in lower tiers or who are subscribed through a base “no fee” trial plan (e.g., a free **30**-day trial period). Illustratively, a user paying the subscription fee will be able to access higher-level security features (e.g., two-factor authentication), more advanced shipping/transportation options and a wider variety of sizes for the smart moving container **100**. The user subscription module **408** will also provide for the creation and maintenance of individual user profiles for each subscribed user in conjunction with the user profile module **410** that will manage and administer such user profiles. Further, in an embodiment, the graphical user interface module **404**, data display interface module **418**, and the communications module **412** are used to facilitate the input/output and display of electronic data and other information (e.g., a graphical user interface) to, illustratively, the users employing the data processing system **300** (e.g., a touch screen) and executing the smart moving container app **400**. The data collection module **406** facilitates smart moving container services information collection from the plurality of users. The data collection module **406** may also be used to collect a variety of smart moving container services information from other virtual and/or electronic sources accessible via the well-known Internet and individual third-party websites hosted thereon. The operations executed by each and every of the foregoing modules are, for example, as discussed throughout this disclosure.

[0063] Those skilled in the art will appreciate that the present disclosure contemplates the use of systems configurations and/or computer instructions that may perform any or all of the operations involved in the smart moving container services herein. The disclosure of computer instructions that include, for example, the smart moving container app **400** and the data processing system **300** instructions is not meant to be limiting in any way. Those skilled in the art will readily appreciate that stored computer instructions and/or systems configurations may be configured in any way while still accomplishing the various goals, features, and advantages according to the present disclosure. The terms “program,” “application,” “software application,” and the like as used herein, are defined as a sequence of instructions designed for execution on a computer system. A “program,” “computer program,” “application,” or “software application” may include a subroutine, a function, a procedure, an object method, an object implementation, an executable application, an applet, a servlet, a source code, an object code, a shared library/dynamic load library, and/or other sequence of instructions designed for execution on a computer system. Accordingly, the applications herein may be written using any number of programming languages and/or executed on compatible platforms including, but not limited to, JavaScript®, PHP (PHP:

Hypertext Preprocessor), WordPress®, Drupal, Laravel®, React.js, Angular.js, and Vue.js®. Computer readable program instructions for carrying out operations of the disclosed embodiments may be assembler instructions, instruction-set-architecture (ISA) instructions, machine instructions, machine dependent instructions, microcode, firmware instructions, state-setting data, or either source code or object code written in any combination of one or more programming languages, including an object oriented programming language such as Java, Smalltalk, C++ or the like, and conventional procedural programming languages, such as the “C” programming language or similar programming languages. The computer readable program instructions may execute entirely on one or more standalone computers, partly on one or more standalone computers, as a stand-alone software package, partly on one or more standalone computers and partly on one or more remote computers, partly on one or more standalone computers and partly on one or more distributed computing environments (such as a cloud environment), partly on one or more remote computers and partly on one or more distributed computing environments, entirely on one or more remote computers or servers, or entirely on one or more distributed computing environments. Standalone computers, remote computers, and distributed computing environments may be connected to each other through any type of network or combination of networks, including local area networks (LANs), wide area networks (WANs), through the Internet (e.g., using an Internet Service Provider), or the connection may be made to external computers.

[0064] Turning our attention to FIG. 8, a flowchart of illustrative operations **500** is shown for user onboarding and subscription in accordance with an embodiment. As noted above, the disclosed embodiments herein contemplate a subscription-based service for use by the users and may also be used in a no-fee or non-subscribed platform. As such, at step **502**, receiving a user request. At step **504**, determining if a user is a new or returning user and, if a new user, then at step **506** determining whether the user is a subscriber or not (e.g., in accordance with a user agreement and/or user privacy/security agreement). If the new user agrees to subscribe, then the new user is subscribed and, at step **508**, a user profile is created and the user profile is stored, at step **510**. Turning our attention briefly to FIG. 9, an illustrative user profile **600** is shown in accordance with an embodiment. The illustrative user profile **600** may comprise username **602**, user identification (ID) **604** (as issued during the subscription operation, as detailed above), smart moving container inventory information **606**, registered user devices **608**, registered sensors **610**, shipping history **612**, financial information **616** (e.g., bank routing and account information), and other information **614**.

[0065] Turning our attention back to FIG. 8, if the user declines to become a subscriber, then they may proceed as a non-subscriber and will be entering, at step **512**, a temporary user profile and storing the user profile at step **514**. Illustratively, such a temporary user profile may be an abbreviated version of the aforementioned subscriber's user profile. If the user is a returning user (or one of the newly subscribed users or new non-subscribed user), their identity is authenticated, at step **516** (e.g., using the aforementioned user ID). In an embodiment, two-factor authentication is used to verify the user's credentials (e.g., a text message to the data processing system **300** associated with this user). If not authenticated, then the operations end at step **518**, and if they are authenticated then, at step **520**, their respective user profile is retrieved. A determination is made, at step **522**, as to whether any updates to the user profile are desired and if so, then at step **524**, receiving the update, and updating and storing the user profile, at step **526**. If no updates are necessary (or after the updating operations are completed), at step **528**, there is an enablement of location-based services (e.g., on the data processing system **300** associated with the user) to facilitate the location-based services operations detailed herein.

[0066] Turning our attention to FIG. 10, a flowchart of operations **700** is shown for using a smart moving container **100** for tracking the contents therein and the location thereof and also monitors and/or mitigates factors that impact the transportation of goods as related to damage control, weight, temperature, humidity, impact detection, and inventory management in accordance with an

embodiment. More particularly, at step **702**, attaching one or more location tracking devices capable of tracking a real-time location of the smart moving container, at step **704** attaching one or more temperature sensors capable of monitoring and transmitting a real-time temperature of contents within the smart moving container, at step **706**, attaching one or more weight sensors capable of monitoring and transmitting a real-time weight of the smart moving container, at step **708**, attaching one or more impact sensors capable of monitoring for and transmitting an alert when a real-time impact or shock is applied to the smart moving container, at step **708**, and at step **710**, attaching one or more humidity sensors capable of monitoring humidity. Optionally, in an embodiment, the aforementioned sensors may be registered with a central server (e.g., the server **260**). At step **712**, applying specialized padding to the smart moving container and at step **714** assigned a unique inventory tracking identification code to the smart moving container. Then, at step **716**, monitoring a location, temperature, weight, impact, humidity and inventory status of the smart moving container using the one or more location tracking devices, the one or more temperature sensors, the one or more humidity sensor, and the inventory tracking identification code, and as step **718**, transmitting, the location, temperature, weight, impact, humidity, and inventory status of the smart moving container to at least one user device.

[0067] Devices or system modules that are in at least general communication with each other need not be in continuous communication with each other, unless expressly specified otherwise. In addition, devices or system modules that are in at least general communication with each other may communicate directly or indirectly through one or more intermediaries. Moreover, it is understood that any system components described or named in any embodiment or claimed herein may be grouped or sub-grouped (and accordingly implicitly renamed) in any combination or sub-combination as those skilled in the art can imagine as suitable for the particular application and still be within the scope and spirit of the claimed embodiments of the present invention. For an example of what this means, if the invention was a controller of a motor and a valve and the embodiments and claims articulated those components as being separately grouped and connected, applying the foregoing would mean that such an invention and claims would also implicitly cover the valve being grouped inside the motor and the controller being a remote controller with no direct physical connection to the motor or internalized valve, as such the claimed invention is contemplated to cover all ways of grouping and/or adding of intermediate components or systems that still substantially achieve the intended result of the invention. A description of an embodiment with several components in communication with each other does not imply that all such components are required. On the contrary, a variety of optional components are described to illustrate the wide variety of possible embodiments of the present invention.

[0068] As is well known to those skilled in the art, many careful considerations and compromises typically must be made when designing for the optimal manufacture of a commercial implementation of any system, and in particular, the embodiments of the present invention. A commercial implementation in accordance with the spirit and teachings of the present invention may be configured according to the needs of the particular application, whereby any aspect(s), feature(s), function(s), result(s), component(s), approach(es), or step(s) of the teachings related to any described embodiment of the present invention may be suitably omitted, included, adapted, mixed and matched, or improved and/or optimized by those skilled in the art, using their average skills and known techniques, to achieve the desired implementation that addresses the needs of the particular application.

[0069] Those of skill in the art will appreciate that where appropriate, some embodiments of the disclosure may be practiced in network computing environments with many types of computer system configurations, including personal computers, hand-held devices, multi-processor systems, microprocessor-based or programmable consumer electronics, network PCs, minicomputers, mainframe computers, and the like. Where appropriate, embodiments may also be practiced in distributed computing environments where tasks are performed by local and remote processing

devices that are linked (either by hardwired links, wireless links, or by a combination thereof) through a communications network. In a distributed computing environment, program modules may be located in both local and remote memory storage devices. "Software" may refer to prescribed rules to operate a computer. Examples of software may include code segments in one or more computer-readable languages; graphical and/or textual instructions; applets; pre-compiled code; interpreted code; compiled code; and computer programs. A network is a collection of links and nodes (e.g., multiple computers and/or other devices connected together) arranged so that information may be passed from one part of the network to another over multiple links and through various nodes. Examples of networks include the Internet, the public switched telephone network, wireless communications networks, computer networks (e.g., an intranet, an extranet, a local-area network, or a wide-area network), wired networks, and wireless networks.

[0070] Aspects of the present invention are described herein with reference to flowchart illustrations and/or block diagrams of methods, apparatus (systems) and computer program products according to embodiments of the invention. It will be understood that each block of the flowchart illustrations and/or block diagrams, and combinations of blocks in the flowchart illustrations and/or block diagrams, can be implemented by computer program instructions. These computer program instructions may be provided to a processor of a general-purpose computer, special purpose computer, or other programmable data processing apparatus to produce a machine, such that the instructions, which execute via the processor of the computer or other programmable data processing apparatus, create means for implementing the functions/acts specified in the flowchart and/or block diagram block or blocks. The flowchart and block diagrams in the figures illustrate the architecture, functionality, and operation of possible implementations of systems, methods and computer program products according to various embodiments. In this regard, each block in the flowchart or block diagrams may represent a module, segment, or portion of code, which comprises one or more executable instructions for implementing the specified logical function(s). It should also be noted that, in some alternative implementations, the functions noted in the block may occur out of the order noted in the figures. For example, two blocks shown in succession may, in fact, be executed substantially concurrently, or the blocks may sometimes be executed in the reverse order, depending upon the functionality involved. It will also be noted that each block of the block diagrams and/or flowchart illustration, and combinations of blocks in the block diagrams and/or flowchart illustration, can be implemented by special purpose hardware-based systems that perform the specified functions or acts, or combinations of special purpose hardware and computer instructions.

[0071] These computer program instructions may also be stored in a computer readable medium that can direct a computer, other programmable data processing apparatus, or other device to function in a particular manner, such that the instructions stored in the computer readable medium produce an article of manufacture including instructions which implement the function/act specified in the flowchart and/or block diagram block or blocks. Further, although process steps, method steps, algorithms or the like may be described in a sequential order, such processes, methods and algorithms may be configured to work in alternate orders. In other words, any sequence or order of steps that may be described does not necessarily indicate a requirement that the steps be performed in that order. The steps of processes described herein may be performed in any order practical. Further, some steps may be performed simultaneously.

[0072] It will be readily apparent that the various methods and algorithms described herein may be implemented by, e.g., appropriately programmed general purpose computers and computing devices. Typically, a processor (e.g., a microprocessor) will receive instructions from a memory or like device, and execute those instructions, thereby performing a process defined by those instructions. Further, programs that implement such methods and algorithms may be stored and transmitted using a variety of known media. When a single device or article is described herein, it will be readily apparent that more than one device/article (whether or not they cooperate) may be

used in place of a single device/article. Similarly, where more than one device or article is described herein (whether or not they cooperate), it will be readily apparent that a single device/article may be used in place of the more than one device or article. The functionality and/or the features of a device may be alternatively embodied by one or more other devices which are not explicitly described as having such functionality/features. Thus, other embodiments of the present invention need not include the device itself.

[0073] The term “computer-readable medium” as used herein refers to any medium that participates in providing data (e.g., instructions) which may be read by a computer, a processor or a like device. Such a medium may take many forms, including but not limited to, non-transitory, non-volatile media, volatile media, and transmission media. Non-volatile media include, for example, optical or magnetic disks and other persistent memory. Volatile media include dynamic random-access memory (DRAM), which typically constitutes the main memory. Transmission media include coaxial cables, copper wire and fiber optics, including the wires that comprise a system bus coupled to the processor. Transmission media may include or convey acoustic waves, light waves and electromagnetic emissions, such as those generated during radio frequency (RF) and infrared (IR) data communications. Common forms of computer-readable media include, for example, a floppy disk, a flexible disk, hard disk, magnetic tape, any other magnetic medium, a CD-ROM, DVD, any other optical medium, a RAM, a PROM, an EPROM, a FLASH-EEPROM, removable media, flash memory, a “memory stick”, any other memory chip or cartridge, a carrier wave as described hereinafter, or any other medium from which a computer can read. Various forms of computer readable media may be involved in carrying sequences of instructions to a processor. For example, sequences of instruction may be delivered from RAM to a processor, may be carried over a wireless transmission medium, and/or may be formatted according to numerous formats, standards or protocols, such as Bluetooth®, 4G, 5G, etc.

[0074] Where databases are described, it will be understood by one of ordinary skill in the art that alternative database structures to those described may be readily employed, and other memory structures besides databases may be readily employed. Any schematic illustrations and accompanying descriptions of any sample databases presented herein are exemplary arrangements for stored representations of information. Any number of other arrangements may be employed besides those suggested by the tables shown. Similarly, any illustrated entries of the databases represent exemplary information only; those skilled in the art will understand that the number and content of the entries can be different from those illustrated herein. Further, despite any depiction of the databases as tables, an object-based model could be used to store and manipulate the data types of the present invention and likewise, object methods or behaviors can be used to implement the processes of the present invention.

[0075] A “computer system” may refer to a system having one or more computers, where each computer may include a non-transitory computer-readable medium embodying software to operate the computer or one or more of its components. Examples of a computer system may include: a distributed computer system for processing information via computer systems linked by a network; two or more computer systems connected together via a network for transmitting and/or receiving information between the computer systems; a computer system including two or more processors within a single computer; and one or more apparatuses and/or one or more systems that may accept data, may process data in accordance with one or more stored software programs, may generate results, and typically may include input, output, storage, arithmetic, logic, and control units. A “network” may refer to a number of computers and associated devices that may be connected by communication facilities. A network may involve permanent connections such as cables or temporary connections such as those made through the telephone or other communication links. A network may further include hard-wired connections (e.g., coaxial cable, twisted pair, optical fiber, waveguides, etc.) and/or wireless connections (e.g., radio frequency waveforms, free space optical waveforms, acoustic waveforms, etc.). Examples of a network may include: an internet, such as the

Internet; an intranet; a LAN; a wide area network (WAN); and a combination of networks, such as an internet and an intranet.

[0076] As noted above, in some embodiments the method or methods described above may be executed or carried out by a computing system including a non-transitory computer-readable storage medium, also described herein as a storage machine, that holds machine-readable instructions executable by a logic machine (i.e., a processor or programmable control device) to provide, implement, perform, and/or enact the above described methods, processes and/or tasks. When such methods and processes are implemented, the state of the storage machine may be changed to hold different data. For example, the storage machine may include memory devices such as various hard disk drives, CD, or DVD devices. The logic machine may execute machine-readable instructions via one or more physical information and/or logic processing devices. For example, the logic machine may be configured to execute instructions to perform tasks for a computer program. The logic machine may include one or more processors to execute the machine-readable instructions. The computing system may include a display subsystem to display a GUI, or any visual element of the methods or processes described above. For example, the display subsystem, storage machine, and logic machine may be integrated such that the above method may be executed while visual elements of the disclosed system and/or method are displayed on a display screen for user consumption. The computing system may include an input subsystem that receives user input. The input subsystem may be configured to connect to and receive input from devices such as a mouse, keyboard, or gaming controller. For example, a user input may indicate a request that a certain task is to be executed by the computing system, such as requesting the computing system to display any of the above-described information or requesting that the user input updates or modifies existing stored information for processing. A communication subsystem may allow the methods described above to be executed or provided over a computer network. For example, the communication subsystem may be configured to enable the computing system to communicate with a plurality of personal computing devices. The communication subsystem may include wired and/or wireless communication devices to facilitate networked communication. The described methods or processes may be executed, provided, or implemented for a user or one or more computing devices via a computer-program product such as via an application programming interface (API).

[0077] Thus, the steps of the disclosed method(s) and the associated discussion herein above can be defined by the computer program instructions stored in a memory and/or data storage device and controlled by a processor executing the computer program instructions. Accordingly, by executing the computer program instructions, the processor executes an algorithm defined by the disclosed method. For example, the computer program instructions can be implemented as computer executable code programmed by one skilled in the art to perform the illustrative operations defined by the disclosed methods. Further, it will be appreciated that any flowcharts, flow diagrams, state transition diagrams, pseudo code, program code and the like represent various processes which may be substantially represented in computer readable medium and so executed by a computer, machine, or processor, whether or not such computer, machine or processor is explicitly shown. One skilled in the art will recognize that an implementation of an actual computer or computer system may have other structures and may contain other components as well, and that a high-level representation of some of the components of such a computer is for illustrative purposes.

[0078] It should be understood, of course, that the foregoing relates to exemplary embodiments of the invention and that modifications may be made without departing from the spirit and scope of the invention as set forth in the following claims.

Claims

- 1.** A smart moving container comprising: a container body having a base and a plurality of sidewalls joined to the base, the base and the plurality of sidewalls forming a body cavity therebetween, the plurality of sidewalls having an interior wall surface and an exterior wall surface; a container lid comprising one or more portions; a connection system operative to connect the container lid to the container body; one or more location tracking devices capable of tracking and transmitting a real-time location of the smart moving container; one or more weight sensors capable of monitoring and transmitting a real-time weight within the body cavity; and one or more impact sensors capable of monitoring for a real-time impact or shock and transmitting an impact alert when the real-time impact or shock is applied to the smart moving container.
 - 2.** The smart moving container of claim 1, wherein the container body and the container lid are formed a material selected from the group consisting of high-density (HD) polyethylene and polypropylene.
 - 3.** The smart moving container of claim 1, wherein the container lid comprises a first hinged portion and a second hinged portion.
 - 4.** The smart moving container of claim 1, wherein the connection system comprises one or more hinges hingedly connecting the one or more portions of the container lid to the container body.
 - 5.** The smart moving container of claim 1, wherein the container body comprises one or more handles.
 - 6.** The smart moving container of claim 1, wherein the one or more location tracking devices are selected from the group consisting of a radio-frequency identification (RFID) tag, a barcode, a short-range wireless technology tracking device, an ultra-wideband technology tracking device, and a combination thereof.
 - 7.** The smart moving container of claim 1, wherein the one or more location tracking devices, the one or more weight sensors, and the one or more impact sensors are located within the body cavity.
 - 8.** The smart moving container of claim 1, wherein at least one of the one or more location tracking devices, the one or more weight sensors, and the one or more impact sensors is mounted on one sidewall of the plurality of sidewalls.
 - 9.** The smart moving container of claim 1, wherein the container lid is configured to enable the smart moving container to be stacked.
 - 10.** The smart moving container of claim 1, comprising at least one light emitting diode mounted on the interior wall surface.
 - 11.** The smart moving container of claim 1, further comprising padding capable of protecting contents housed within the body cavity.
 - 12.** The smart moving container of claim 1, further comprising one or more temperature sensors capable of monitoring and transmitting a real-time temperature within the body cavity.
 - 13.** The smart moving container of claim 1, further comprising one or more humidity sensors capable of monitoring and transmitting a real-time humidity exposure level within the body cavity.
 - 14.** A method of tracking, monitoring, and protecting goods during transportation, comprising: providing the smart moving container of claim 1; configuring the one or more location tracking devices to track the real-time location; configuring the one or more weight sensors to monitor and transmit the real-time weight; configuring the one or more impact sensors to monitor for the real-time impact or shock and to transmit the impact alert when the real-time impact or shock is applied; monitoring the real-time location, the real-time weight, and the impact alert of the smart moving container using the one or more location tracking devices, the one or more weight sensors, and the one or more impact sensors; and transmitting the real-time location, the real-time weight, and the impact alert to at least one user device.
-